

PATHUM RATHNAYAKA

SOFTWARE ENGINEER

[Linkedin](#) [Portfolio](#) [GitHub](#)

• thilinapathumrathnayaka@gmail.com • +94 76 1040031 • Arthur v. Dias Road, Panadura

SUMMARY

Software Engineer with experience building real-world software solutions, including HR & Payroll systems, SaaS platforms, and business websites. Enjoy working across different stages of software development, from designing features to implementing and maintaining production applications. Comfortable working both independently and as part of a team, with a focus on delivering reliable solutions that solve real business problems.

EDUCATION

BSc (Hons) Computer Science University of Bolton	Present
Graduate Diploma in Software Engineering Institute of Software Engineering – IJSE	2023 - 2025
G.C.E. Advanced Level (Technology stream) Thambuththegama national college	2019 - 2021

PROFESSIONAL EXPERIENCE

Intern Software Engineer

Cenzios (PVT) Ltd | Sri Lanka | Remote | Sep 2025 – Mar 2026

- Served as the primary developer for CenzHRM, a multi-tenant SaaS Human Resource Management platform, leading the end-to-end development of both frontend and backend systems, including the administrative portal and core business modules.
- Designed and implemented scalable frontend applications using React, Next.js, TypeScript, Redux Toolkit, React Query, Tailwind CSS, and Shadcn/ui, delivering a responsive and user-friendly experience across devices.
- Developed backend services and RESTful APIs using Node.js, Express.js, PostgreSQL, and MongoDB, supporting employee management, attendance tracking, leave management, payroll operations, and role-based access control.
- Contributed to architectural discussions and requirement analysis for the Voyage tourism booking platform, gaining practical exposure to NestJS-based backend architecture, authentication flows, database design, and scalable SaaS development practices.
- Refactored and enhanced core features of the SNP Holdings website using React, Next.js, and Node.js, improving maintainability, performance, and overall user experience.
- Collaborated in Agile/Scrum development processes using Jira for sprint planning, task management, bug tracking, and progress monitoring, ensuring timely delivery of project milestones.

Freelance Software Engineer

Upwork | Part time | Jan 2024 – Present

- Developed custom web solutions, dashboards, and business management systems tailored to client requirements.
- Fixed critical bugs, optimized application performance, and enhanced user experience for existing production systems.
- Integrated third-party services, payment gateways, authentication providers, and external APIs to extend application functionality.

PROJECTS

Marketplace SaaS Platform (Enterprise Microservices)

- Architected and developed a cloud-ready Marketplace SaaS platform based on a microservices architecture, implementing secure authentication and authorization, distributed business services, scalable inter-service communication, and optimized data management to support high-availability enterprise applications.

Github : <https://github.com/PathumRathnayaka/Marketplace-Microservice.git>

Technologies: Java, Spring Boot, PostgreSQL, RabbitMQ, JWT, REST APIs, API Gateway, Maven,

CenzHRM – Human Resource & Payroll Management System (Industrial Project)

- During my Software Engineering internship, developed CenzHRM, an HR and Payroll platform used by multiple organizations, implementing employee management, payroll, salary calculations (EPF/ETF), loans, attendance tracking, and role-based access control for a live production environment.

Web : <https://payroll.dev.cenzios.com>

Technologies: React.js, Next.js, TypeScript, Tailwind CSS, Redux, MySQL, REST APIs,

Hybrid POS System (Offline-First Point of Sale Platform)

- Developed a hybrid offline-first Point of Sale (POS) system for a local business client using JavaFX and Spring Boot, supporting seamless online and offline operations with automated synchronization to a cloud-based microservices architecture via an API Gateway.
- The platform provides end-to-end POS functionality, including sales, inventory, product, customer, supplier, purchasing, and reporting management, with reliable data synchronization and scalable cloud integration.

Github : <https://github.com/PathumRathnayaka/pos.git>

Technologies: Java, JavaFX, Spring Boot, MySQL, REST APIs, API Gateway, Maven,

SNP Holdings Corporate Website

- Developed the corporate website for SNP Holdings, a leading Sri Lankan provider of sublimation printing, DTF printing, and sportswear manufacturing, using Next.js. Built responsive user interfaces, optimized website performance and SEO, and delivered a modern digital platform to showcase the company's products and services.

Web : <https://snpholdings.online>

Technologies: Next.js, TypeScript, Tailwind CSS

AI Employee Productivity Monitoring System

- A real-time monitoring system using Python, OpenCV, MediaPipe, YOLO, Flask, and React to detect phone usage, sleeping, and away-from-desk behavior through webcam input. Activity events are logged to MongoDB and streamed live to a React dashboard using WebSockets, providing instant status updates and daily productivity summaries.

Github : <https://github.com/PathumRathnayaka/AI-powered-productivity-monitoring-system.git>

Technologies: Python, OpenCV, MediaPipe, YOLO, Flask, MongoDB, and React

TECHNICAL SKILLS

- | | |
|------------------|---|
| • Backend | Java Spring Boot, Spring WebFlux, Spring Security, Spring Cloud Gateway, Node.js (Express.js, NestJS), Python, REST APIs, GraphQL, Kafka, RabbitMQ, JWT, Auth0 |
| • Database | MySQL, PostgreSQL, MongoDB, Redis, Azure SQL, Supabase, Prisma ORM, Hibernate |
| • Frontend | React.js, Next.js, TypeScript, Tailwind CSS, Redux |
| • Cloud & DevOps | Microsoft Azure (App Service, AKS, Azure DevOps, Blob Storage, Azure SQL, Key Vault, Monitor), AWS (EC2, RDS, S3, Lambda, Amplify, CloudFront), GCP, Docker, Kubernetes, Nginx, Jenkins, GitHub Actions |
| • Cloud & DevOps | Microservices, SaaS Architecture, API Gateway, Datadog, CI/CD Pipelines, Infrastructure Automation |